

Hydrogenic Bound States from Phase Holonomy, Part I

Corrected Effective-Sector Consistency Note

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Abstract

This note asks a deliberately limited question: if a Super Information Theory (SIT) effective description is specified to contain the ordinary locally $U(1)$ -invariant Maxwell and charged-matter sector in a constant-background limit, is standard hydrogen compatible with that limit? The answer is yes by construction. A localized charge obeys the assumed Maxwell Gauss law, and the assumed Schrodinger or Dirac Hamiltonian supplies the standard bound-state spectrum. Connection and holonomy language provides a compatible geometric description, but it does not independently derive QED, electric charge, the Coulomb normalization, or hydrogenic quantization from SIT.

The published paper's conservative core survives, but three technical claims are corrected. First, for a spatially varying gauge kinetic function $f_2(r)$, the radial potential is the nonlocal integral

$$\phi(r) = \int_r^\infty \frac{Q}{4\pi f_2(s)s^2} ds. \quad (1)$$

not generally $Q/[4\pi r f_2(r)]$. Second, the fractional energy relation $\delta E_n/E_n \approx -2\delta f_2/f_2$ applies directly only to a spatially constant rescaling under a declared charge normalization; varying profiles require state-dependent expectation values. A new dimensionless $1s/2s$ example now converts those level corrections into the transition difference that an experiment would actually observe and reproduces it by a separate arbitrary-precision quadrature path. This is an arithmetic demonstration, not a prediction or bound. Third, phase degree, Wilson-loop holonomy, and gauge-covariant circulation are separated. The corrected Part II is a conditional Bohr reconstruction plus obstruction analysis and no longer supports a claim of necessary SIT hydrogen structure. Neither paper validates the other.

Scope and evidence boundary

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1. Status of the Result

The phrase *recovery* has two possible meanings. A theory can dynamically derive a low-energy sector from deeper equations, or a model can declare that sector as its limiting action and verify internal consistency. This paper does the latter. It assumes an effective action containing Maxwell and charged matter when informational backgrounds are constant. Standard hydrogen then follows from standard equations.

That is a useful model-building constraint: any proposed extension must reduce to established atomic physics in its domain. It is not evidence that the deeper SIT fields generate the gauge group, matter representations,

charge spectrum, electron and proton masses, or QED radiative structure. The title is retained for DOI continuity, but *derivation within SIT* is replaced by *effective-sector consistency note*.

The revision distinguishes three statuses:

Statement	Status
Standard Maxwell plus Schrodinger / Dirac dynamics yields standard hydrogen	assumed-sector recovery
Connection / holonomy language is compatible with quantum phase transport	geometric restatement with convention limits
SIT informational fields dynamically produce QED and its parameters	open, not derived here

2. Declared Effective Action

Use the gauge convention

$$\begin{aligned}\mathcal{L}_{\text{EM}} &= -\frac{1}{4}f_2(x)F_{\mu\nu}F^{\mu\nu}, \\ D_\mu &= \partial_\mu + \frac{iq}{\hbar}A_\mu.\end{aligned}\tag{2}$$

with a separately declared charged-matter Lagrangian. The function f_2 is dimensionless and positive in the admitted regime. The constant-background branch sets $f_2=f_2>0$; a field redefinition and charge convention then determine the measured electromagnetic coupling. What is held fixed during a variation of f_2 must be stated. Holding a bare q fixed is not the same as holding the measured fine-structure constant fixed.

No microscopic formula for $f_2(\rho_t, R_{\text{coh}})$ is derived here. Coefficients such as ϵ and η remain phenomenological. Calling the expansion an EFT does not establish scale separation, operator completeness, radiative stability, or constraints; those require a full action and matching program.

2.1 Normalization choices

The coefficient in front of F^2 and the symbol called electric charge are not separately observable without a normalization convention. For constant f_2 , one may canonically normalize the gauge field and move the constant factor into the matter coupling. For varying $f_2(x)$, that local redefinition introduces additional derivative structure and cannot be treated as a global change of units. A revision must therefore state whether Q is a bare source charge, a canonically normalized low-energy charge, or a measured asymptotic charge.

The current calculation fixes the action convention, treats Q and the test charge as declared parameters, and asks only how a specified radial f_2 modifies the classical potential in that convention. It does not claim that ϵ or η is invariant under every field redefinition. Any comparison with a bound on the fine-structure constant must first translate to the observable coupling in the same frame and normalization.

3. Correct Static Radial Field

For a static point source and spherical $f_2(r)$, variation of the declared gauge action gives, outside the source,

$$4\pi r^2 f_2(r) E_r(r) = Q.\tag{3}$$

With $E_r = -d\phi/dr$ and $\phi(\infty) = 0$,

$$\begin{aligned} E_r(r) &= \frac{Q}{4\pi f_2(r)r^2}, \\ \phi(r) &= \int_r^\infty \frac{Q}{4\pi f_2(s)s^2} ds. \end{aligned} \tag{4}$$

For $f_2 = f_0$, this reduces to $Q/(4\pi f_0 r)$. If $f_2(r) = f_0[1 + \delta g(r)]$ with small δ , then

$$\phi(r) = \frac{Q}{4\pi f_0} \left[\frac{1}{r} - \delta \int_r^\infty \frac{g(s)}{s^2} ds \right] + \mathcal{O}(\delta^2). \tag{5}$$

The published local expression replaces the integral by $g(r)/r$. That is exact for constant g , but not for a general profile. A slow-variation approximation needs a quantitative derivative bound over the radial support relevant to the state.

The package uses $g(r) = L/(r+L)$, for which

$$\int_r^\infty \frac{g(s)}{s^2} ds = \frac{1}{r} + \frac{1}{L} \log\left(\frac{r}{r+L}\right). \tag{6}$$

The deterministic script confirms this analytic expression by numerical quadrature and shows that it differs from $g(r)/r$ at a declared sample radius.

4. Energy Shifts

The perturbing potential energy for a particle of charge q_{test} is $\delta V = q_{\text{test}} \delta \phi$. First-order stationary perturbation theory gives

$$\delta E_{n\ell m} = \langle n\ell m | \delta V | n\ell m \rangle. \tag{7}$$

The result is state dependent whenever $g(r)$ varies. It can split states that were degenerate in an oversimplified Coulomb treatment, depending on the profile and operator. A single local value of δf_2 cannot replace the wavefunction expectation.

For a spatially constant $f_2 = f_0(1 + \delta)$ under fixed source / test charge conventions, the Coulomb coefficient rescales by $(1 + \delta)^{-1}$. In the nonrelativistic Coulomb model, energies scale as the square of that coefficient:

$$\begin{aligned} \frac{E_n(\delta)}{E_n(0)} &= (1 + \delta)^{-2}, \\ \frac{\delta E_n}{E_n} &= -2\delta + \mathcal{O}(\delta^2). \end{aligned} \tag{8}$$

This is the legitimate origin of the published -2δ expression. It is a constant-branch scaling identity, not a general spatial-profile result. The script computes illustrative 1s and 2s expectations in dimensionless units to make the distinction reproducible. It does not fit data or constrain SIT.

4.1 Domain of first-order perturbation theory

The expectation-value formula assumes that the unperturbed Hamiltonian is self-adjoint on its stated domain, that the perturbation is sufficiently small for the selected state, and that degeneracies are handled with the appropriate matrix problem. A spatially varying central profile preserves rotational symmetry but can alter radial expectation values by n and 1 . A noncentral coherence profile can also mix magnetic sublevels and requires a different calculation.

The illustrative script uses the nonrelativistic radial probabilities

$$\begin{aligned} P_{1s}(r) &= 4r^2 e^{-2r}, \\ P_{2s}(r) &= \frac{r^2(2-r)^2 e^{-r}}{8}. \end{aligned} \tag{9}$$

in units where $a_0=1$. It verifies their normalization over the declared numerical domain, truncates the main radial integral at a stated upper bound, and starts at a small positive radius to avoid evaluating intermediate expressions at zero. Convergence of that numerical quadrature is an internal arithmetic check, not a theory uncertainty. A publishable atomic calculation would need relativistic and radiative terms at the precision relevant to the proposed effect, as well as finite nuclear size, recoil, and experimental line-shape conventions.

4.2 Illustrative 1s-2s transition calculation

For the package profile $g(r)=L/(r+L)$ with $L=a_0$, $\text{delta}=0.01$, and atomic units $a_0=K=1$, the finite-domain Simpson calculation gives

$$\begin{aligned} \langle I_g \rangle_{1s} &= 0.3683085177, & \langle I_g \rangle_{2s} &= 0.05006723114, \\ \delta E_{1s} &= 0.003683085177 E_h, & \delta E_{2s} &= 0.0005006723114 E_h. \end{aligned} \tag{10}$$

Since the unperturbed nonrelativistic transition is $E_2-E_1=3/8$ hartree, the computed transition correction is

$$\begin{aligned} \delta(E_2 - E_1) &= \delta E_{2s} - \delta E_{1s} = -0.003182412865 E_h, \\ \frac{\delta \nu}{\nu} &= -0.008486434307. \end{aligned} \tag{11}$$

The sign follows from the selected positive profile and fixed-charge convention: the less compact 2s state samples a smaller profile-weighted integral than 1s. These values are deliberately dimensionless and illustrative. They cannot be multiplied into the measured hydrogen frequency and called an SIT prediction because the example supplies no physical delta , field source, screening law, relativistic matching, or uncertainty model.

A separate `mpmath` implementation uses adaptive 60-decimal quadrature on $[0, 1, 4, 12, \text{infinity}]$, rather than the package's fixed-grid Simpson routine. It returns 0.3683085377 , 0.05006723365 , and a fractional transition correction of -0.008486434773 , agreeing within the declared finite-domain tolerance. This is an internally authored independent numerical pathway, not independent external review or experimental reproduction.

5. Holonomy and Quantization

For a nonvanishing complex section in a chosen local trivialization, the phase degree around a loop is

$$\oint_{\gamma} \nabla \theta \cdot d\ell = 2\pi w, \quad w \in \mathbb{Z}. \quad (12)$$

For the stated covariant derivative, the gauge-invariant circulation is

$$\oint_{\gamma} \left[\nabla \theta - \frac{q}{\hbar} A \right] \cdot d\ell = 2\pi w - \frac{q}{\hbar} \Phi. \quad (13)$$

It equals an integer multiple of 2π only when the Abelian Wilson loop is trivial, $(q/\hbar)\Phi \in 2\pi\mathbb{Z}$. Global single-valuedness of a quantum state does not require every Wilson loop to be trivial. The Aharonov-Bohm effect is a standard counterexample to that stronger statement.

Bohr-Sommerfeld quantization is a semiclassical action condition. It is not the modern derivation of the full hydrogen spectrum and should not identify the principal quantum number with orbital angular momentum. In this paper the Schrodinger or Dirac eigenvalue problem is assumed in the recovered sector; the holonomy language is interpretive, not a replacement spectral derivation.

6. Part I and Corrected Part II

The published Part I assigns Part II a necessity result for discrete hydrogenic structure. Corrected Part II retracts that result. Its retained calculation assumes a Coulomb potential, circular relative-coordinate mechanics, and the historical $L=N\hbar$ condition, then reconstructs Bohr scaling. It also records that a point removed from three-dimensional space does not create the loop topology claimed for a proton-centered winding sector and that winding does not set a radial scale.

The cross-paper contract is therefore:

- Part I assumes the conventional effective gauge and quantum sectors and checks their consistency with the declared SIT limit.
- Part II retains a conditional classical Bohr reconstruction and several source-specific obstructions.
- Part I does not provide Part II's missing SIT action or source solution.
- Part II does not establish the deeper origin of Part I's assumed sector.
- Neither paper currently constitutes independent validation of the other.

This alignment prevents circular support.

7. Experimental Boundary, Limitations, and Failure Conditions

Hydrogen spectroscopy is sensitive to small changes in electromagnetic couplings, but sensitivity is not a prediction. A testable SIT specialization would require:

1. a derived or independently parameterized $f_2(\rho_t, R_{\text{coh}})$;
2. a source / environment model that fixes ρ_t and R_{coh} across the atom;
3. coefficients and cutoff with units and priors;
4. a wavefunction / operator calculation including standard corrections;
5. transition-frequency differences rather than isolated level slogans;

6. theory and experimental uncertainty; and
7. comparison with existing scalar-coupling and equivalence-principle bounds.

The 1S-2S measurement is a constraint anchor, not validation. Parthey et al. demonstrate the experimental role of systematic corrections in the hydrogen transition measurement [5]. No coefficient bound is reported here because the package contains no mapping from an experimental environment to the SIT fields.

7.1 From a level shift to an observable

Spectroscopy measures transition frequencies, so a common offset to both levels is unobservable in that transition. The relevant quantity is

$$\delta\nu_{ab} = \frac{\delta E_a - \delta E_b}{h}. \quad (14)$$

The source environment, temporal modulation, polarization, field geometry, and state preparation determine whether a proposed profile is common-mode or state selective. If the informational background is externally engineered, ordinary electromagnetic, thermal, collisional, ac-Stark, Zeeman, pressure, and apparatus shifts must be modeled as controls. A correlation with a coherence preparation is not sufficient if that preparation changes standard fields or total energy.

The parameter inference also needs a null model and an uncertainty budget. A reported residual should be converted into a bound on a declared coefficient only after profiling nuisance parameters and accounting for the look-elsewhere effect across candidate profiles. Without this chain, the phrase *falsifiable pathway* names a research program rather than a present prediction.

7.2 Relation to varying-coupling literature

Field-dependent electromagnetic couplings already appear in varying-constant, dilaton, scalar-dark-matter, and equivalence-principle models. Bekenstein gives an early covariant varying-coupling framework [6], while Olive and Pospelov extend that model with dark-matter and cosmological couplings [7]. Damour and Donoghue show why a light scalar requires a composition-dependent coupling map before equivalence-principle limits can be applied [8]. Atomic-clock and spectroscopy papers then make particular scalar couplings operational through specific sensitivity coefficients, source assumptions, and time dependences [9-12].

That literature supplies model structures and constraints on those declared model classes, not support for SIT specifically and not a bound on the dimensionless example above. A future SIT specialization must identify its scalar combination, canonical frame, symmetry, screening behavior, source response, temporal or spatial signature, and map to the same observable couplings used by the comparison experiment. Reusing the form $f(\phi)F^2$ is not itself novel and does not determine a coefficient. Existing atomic-clock or equivalence-principle limits become relevant only after that translation is defined.

8. Citation and Provenance Corrections

The published reference mapping is swapped. Zenodo 17270488 is *Informational Holonomy and the Electron*. Zenodo 17311157 is the coherence-gated Yang-Mills work. Both are same-program lineage and cannot serve as independent validation.

The Zenodo record date is December 21, 2025, while the PDF internally states December 24. Both dates are preserved with their roles. The immutable PDF hash is B4B5A37D8DB47DE97252A015ED5151C89652EC9C1E995402667E33075E0E8B41.

9. Conclusion

Part I survives as a modest consistency note. If the standard electromagnetic and quantum bound-state sectors are assumed as the constant-background limit, standard hydrogen is recovered. This is recovery by construction, not a first-principles SIT derivation.

The varying-dressing potential is an integral, spatially varying shifts require wavefunction expectation values, and integer covariant circulation requires an additional holonomy condition. Corrected Part II no longer supplies necessary hydrogenic topology. The remaining SIT-specific deviations are hypotheses that need a complete model and external constraints. No new spectrum or validated atomic prediction is claimed.

Authorial provenance and research chronology

Artificial intelligence was not involved in authoring or originating the core ideas, scientific theories, hypotheses, or source-based research corrected and developed in this paper. The underlying research program is Micah Blumberg’s original work and forms part of a continuous research history spanning 2011-2026, with central conceptual foundations recorded before ChatGPT’s public release on November 30, 2022.

During later manuscript refinement, AI tools were used in bounded supporting roles including mathematical and computational formulation, code, grammar and spelling, and checks against physics and mathematical literature. These tools did not supply the original theory or authorship. Micah Blumberg selected the claims and sources, directed the correction, and remains responsible for the paper’s arguments, interpretations, and errors.

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